Study of LEAN Methodology in Software Development

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***Abstract*— Lean Methodology was first introduced as a production management and product development process in the early 20th century. By the late 1940s, the Toyota company from Japan was reported to produce automobiles with roughly half the number of labour forces with much faster production rate and higher quality product than the US. [1] The management philosophy of Toyota is given the name 'Lean manufacturing process', which is then studied and integrated into the world of software development. Lean Methodology emphasis on practices that effectively minimise waste in the production development process, while aiming to achieve a higher quality outcome with the lowest possible cost.**

***Keywords*— Lean, waste reduction, methodology, Value Stream Mapping, predictable, robust, continuous delivery.**

I. Introduction

The seven principle of lean software development methodology as stated as follows:

- optimising the whole
- eliminate waste
- build quality in
- learn constantly
- deliver fast
- engage everyone
- keep getting better

Lean software development tends to optimise the whole by focusing on understanding the user requirements and exploring different possibilities of how software can fulfil them within a larger system by the design, development, deployment, and the ability to evolve over time.

Eliminating waste by discarding anything that doesn't directly contribute to customer value such as unnecessary features, and prioritise the quality so time spent fixing defects could be eliminated. Lean methodology requires the examination of the entire development process in order to minimise inefficiencies.

Build quality means to continuously perform testing throughout the integration in the development cycle to detect any problem at an early stage. The application of building smaller units of software means they could be integrated early and often, rather than needing to wait until the end of a development cycle.

Lean development involves learning constantly by continuously exploring multiple available options of the development process, and constantly improving the process after multiple trial and error. It also emphasises building minimum viable products and iterating based on customer feedback, blending both proactive learning and continuous adaptation.

Frequent releases, even daily or continuously to ensure that software is delivered as a steady flow of small changes, rather than large, infrequent updates. This requires robust testing and mistake-proofing mechanisms to ensure quality.

Lean encourages teamwork that includes everyone doing what they are best of. Decision-making is something distributed to the lowest possible level to empower teams and foster teamwork.

Lastly, Lean emphasises continuous improvement at every level of the organisation, adapting practices like XP or Scrum to the specific needs of the team. It encourages using a scientific approach to refine work processes for better outcomes [2].

The seven principles shall be further explored and evaluated in the paper regarding the impact, challenges as well as comparative analysis of Lean methodology with other methodology.

II. Key Practices Of Lean n Software Development

When the software development process occurs in a different organisation, lean practice encourages teamwork. [3] In this section, several popular practices of lean software methodology in software development will be covered such as value stream mapping, kanban and continuous delivery, and how they help in the optimisation of software development process.

A. Value Stream Mapping

Value stream mapping is a lean manufacturing technique that is used to analyse and record a series of events by the current state and design a future state. A value stream map displays high level process steps along with key process data. Method for analysing the current state and designing a future state for the series of events that take a product or service from the beginning of the specific process until it reaches the customer.

B. Continuous Delivery

Continuous delivery, on the other hand, is a study showcasing how small, frequent releases can enhance quality and responsiveness to customer needs. Team produces software in a way that code is constantly released to the production.

C. Kanban

Kanban is applied in software to manage workflow and improve focus on high-priority tasks by limiting work-in-progress (WIP). In a Kanban system, value stream will be mapped on a physical table with columns representing steps of every process. Kanban cards are placed in every column, documenting every state of work. When a work is being completed, move the Kanban token to the next column, representing the workflow of the system development process [2].

Comparing these interpretations reveals that while all approaches aim to maximise value and minimise waste, the application varies by project type and organisational structure, with some projects benefiting more from visual management techniques like kanban, while others focus on automating deployment through continuous delivery to enhance flow.

III. TECHNIQUES OF APPLYING LEAN METHODOLOGY IN SOFTWARE DEVELOPMENT

Lean software development in practice starts by drawing a Value Stream Map. A map or a timeline of the process of customer request solving will be drawn all the way until the problems are solved successfully. The time spent on each activity is marked, as well as how much customer value each activity brings. Waste elimination comes into play, where requirements are developed based on the cycle time of customer request to customer delivery, and time spent to add customer value. The process is iterated numerous times until the most effective way and process to solve the problem is laid out. Delayed commitment means to avoid having a lot of changes in requirements, developers should delay the process of writing requirements in detail, to make it closer to the coding and testing phase and remain all processes as small chunks. The practice of delayed commitment is not to decide until they have the most information that could possibly be known. To have a good design heuristic, irreversible decisions must be made as late as possible, to leave options open for later. The whole practice of Lean is an iterative cycle of particularly 2 to 4 weeks long. Just before the cycle begins, the top priority things that are needed are pieced together into a detailed requirement. A test is written to help understand the requirements, and coding process to pass every test. Every process is documented and integrated into the code base on a daily basis. This way, when an error appears in part of the program and developers stop and fix it, it will not build up a bunch of defects. When something goes wrong, fixing should be done immediately and ensuring every cycle is effective. The whole process is called a rapid cycle [3].

IV. CHALLENGES AND LIMITATIONS IN APPLYING LEAN

A. Delays outside of the Development Group

A software development process is not often done by an independent group, but rather by a bigger team that is trying to change the business process and the software development team will be a part of it that is trying to deliver value to the customer. When developers are faced with different organisational boundaries they tend to have delays in the value stream. Nobody seems to own moving the process forward across the organisational boundaries [3]. Therefore, an environment in the organisation boundary is required to ensure coordination across the whole product or business process and not only the development team.

B. Lean in Large Organisations

Organisations that do not have lean in high levels tend to fall apart. If a company decided to apply Lean methodology, it is better to apply it across a high level application, having Lean as a corporate philosophy, but not all companies want to switch their culture that way. On the other hand, organisations that are in a deeply competitive environment tend to apply lean methodology and once applied, are able to have high quality and constant improvement.

C. Lean Under Contract

All of the Agile Techniques runs into trouble when the organisational boundary that is being crossed is into another company [3]. This is due to the contract signed between the two companies, which requires the exact flow and budget to be listed off before the workflow starts. When a contract pre-defines everything, it is very difficult to have a Lean environment, because a contract will not allow feedback and change after it has been signed.

V. COMPARATIVE ANALYSIS : LEAN AND AGILE

Lean and agile are having a great competition especially among the supply chains and the companies. Supply Chain Management (SCM) has been effective in manufacturing and has become crucial for companies to stay competitive by providing the products according to the customer requirements.

Naylor, Naim and Berry [5] define these two approaches. Lean focuses on reducing waste to create efficient and cost-effective systems while Agile focuses on flexibility and quick responses to changes. This is a comparison of Lean and Agile methods, examining the variations and exploring how to merge them in a supply chain. The summary of differences between Lean and Agile in Table 1.

A. Lean and Agile Manufacturing

The concept of Lean production originated from Toyota Production System (TPS), which aims to eliminate waste including overproduction, inventories, unnecessary processing, unneeded movement of people, unnecessary transport of goods and employee waiting time [7]. Lean production cuts down the resource usage and is connected with “zero inventory” and just-in-time (JIT) approach [6]. However, Lean’s cost advantage enabled it to limit its responsiveness to demand changes, which Agile method was better at handling the availability.

Agile manufacturing prioritises rapid volume and variety adaptability. The foundation of Agile manufacturing is the virtual corporations which are characterised by partner-based networks and making them adaptable to restructure. Christopher [4] suggests that agile can work the best with unpredictable and high product variety while Lean is the most suitable for a stable and predictable environment.

B. Comparison between Lean and Agile

This section examines the differences and similarities between Lean and Agile methodologies, also highlighting the strengths in the market demands.

1) *Market Qualifiers and Winner:* Lean's primary market advantage is cost-efficiency by focusing on minimising expenses to offer products at competitive prices, while Agile achieves in availability by responding quickly to changes in customer requirements. In Lean, customers usually purchase specific products, whereas Agile allows customers to reserve products on a short notice, according to sudden changing demands. Both methodologies prioritise quality and lead time to ensure the delivery time is in high-quality products.

2) *Fashion Good vs. Commodities:* The type of product determines the best methodology for its management. Fashion goods, known for their short life cycles and unpredictable demand, align well with Agile practices, which promote flexibility and quick responses to market changes. This approach allows for rapid iterations to adapt the design to current trends. On the other hand, commodities are well-suited to Lean methodology with their priority on efficiency, waste reduction and predictable scheduling to meet market demands.

3) *Characteristics of Lean and Agile:* Naylor, Naim and Berry [5] compared the characteristics of Lean and Agile methodologies. Both methodologies emphasise the importance of market knowledge and rapid configuration, they pursue the goals differently. Lean focuses on optimising processes by minimising waste and improving efficiency, whereas Agile emphasises rapid configuration to quickly adapt to changing product demands. Additionally, robustness is crucial for Agile manufacturers, enabling them to respond to fluctuations in customers demand, while Lean aims to create smooth demand through forward planning and careful market analysis to minimise waste.

TABLE 1
THE DIFFERENCES BETWEEN LEAN AND AGILE

Aspects	**Lean**	**Agile**
Market Winner	Cost	Availability
Market Qualifiers	Quality and lead time.	Quality and lead time.
Customer Focus	On specific product purchases.	On rapid availability and flexibility.
Ideal Product Type	Commodities with stable demand.	Fashion goods with short life cycles and unpredictable high demand.
Production Approach	Predictable, level scheduling.	Flexible and adaptable to demand changes.
Key Characteristics	Waste reduction, efficiency stability.	Responsiveness, robustness and rapid configuration.
Demand Type	Smooth, predictable demand.	Variable, unpredicted demand.
Best-Suited Environment	High volume, low variety and stable conditions.	High variety, less predictable and dynamic conditions.
Primary Goal	Minimise costs by reducing waste.	Maximise responsiveness and adaptability.

VI. CONCLUSION

Lean Software Development offers a structured approach that prioritises efficiency, responsiveness and quality software delivery. By minimising waste and optimising workflows through specialised techniques, Lean enhances productivity and satisfies the customer expectations. However, challenges like delays and coordination issues might occur, especially in larger or cross-functional environments where robust organisational support is crucial to maintain the workflows. The comparison of Agile and Lean highlights the unique strengths of each and their combination provides a useful framework that enables businesses to maximise their advantages for producing software with high quality in a variety of projects.

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